

Q2

i when n=8	i when n=10
1 ✓	1
2 ✓	2
4 ✓	4
8 ✗	8 ✓
	16 ✗

3 times → 4 times

so we need to deal value

Q $\text{foo}(i=n; i>1; i=i/2)$
 { statement
 }
 ;

$\frac{i}{2}$
 $\frac{n}{2} \Rightarrow \frac{n}{2^1}$
 $\frac{n}{4} \Rightarrow \frac{n}{2^2}$
 $\frac{n}{8} \Rightarrow \frac{n}{2^3}$
 $\frac{n}{16} \Rightarrow \frac{n}{2^4}$
 $\vdots$
 $\frac{n}{2^k}$
 Assume $i < 1$
 $\therefore \frac{n}{2^k} < 1$
 $\frac{n}{2^k} \leq 1$
 $n = 2^k$
 $\log^k = k$
 $O(\log^n)$

Q1 $\text{for}(i=0; i \leq n; i++)$
 { statement
 }

i
 0
 1
 2 → $2 \times 2 < n$
 3 → $3 \times 3 < n$
 4 → $4 \times 4 < n$
 $\vdots$
 $k \rightarrow k \times k < n$
 terminate when $i \geq n$
 $k^2 < n$
 $k < \sqrt{n}$
 $O(\sqrt{n})$

Q2 $\text{for}(i=0; i < n; i++)$ → n
 { statement }
 ;
 $\text{for}(j=0; j < n; j++)$ → n
 { statement }
 ;
 $O(n^2)$
 $O(n)$

Q $p=0$
 $\text{for}(i=1; i < n; i=i \times 2)$

2 $p = p++ \rightarrow \log(n)$

3

for ($j=1; j < p; j=j*2$)

2 statement $\rightarrow \log(p)$

3

$p = \log(n)$
Putting value
 $O(\log \log n)$

Q for ($i=0; i < n; i++$) $\rightarrow n$
 { for ($j=0; j < n; j=j*2$) $\rightarrow n \log(n)$
 2 statement $\rightarrow n \log(n)$
 3

$$\frac{2n \log(n) + n}{O(n \log n)}$$

for ($i=1; i < n; i++$) $O(n)$
 for ($i=1; i < n; i=i+2$) $\frac{n}{2} O(n)$ $\frac{n}{2}$ is also $O(n)$
 for ($i=n; i > 1; i--$) $O(n)$
 for ($i=1; i < n; i=i*2$) $\log_2^n$
 for ($i=1; i < n; i=i*3$) $\log_3^n$
 for ($i=n; i > 1; i=i/2$) $\log_2^n$

Q $i=1$
 $k=1$
 while ($k < n$)
 { statement
 $k = k + i$
 $i = i + 1$

i	k
1	1
2	1+1
3	2+2
4	2+2+3
5	2+2+3+4
...	...
m	2+2+3+4+...+m

Can be written as $\Rightarrow$

for ($i=1, k=1; k < n; i++$)
 { statement
 $k = k + i$
 }

$$= \frac{m(m+1)}{2}$$

 if we take as 1 then

$$\frac{m(m+1)}{2} + 1$$

loop terminates
 $k \geq n$

$$\frac{m^2 + m}{2} = n$$

$$m^2 = n$$

$$m = \sqrt{n}$$

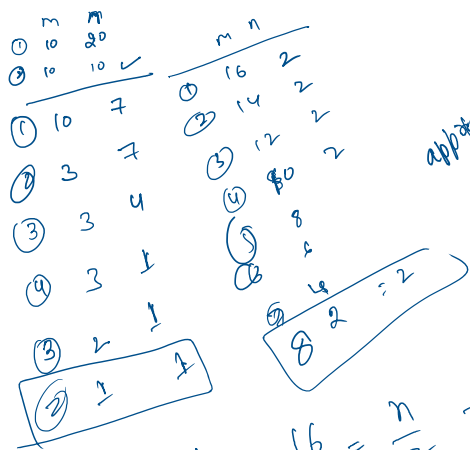
$$O(\sqrt{n})$$

Q

```

while (m < n)
{
    if (m > n)
    {
        m = m - n;
    }
    else
    {
        n = n - m;
    }
}

```



approx $\frac{16}{2}$

$\Rightarrow$

```

for( ; m != n; )
{
    if (m > n)
    {
        m = m - n;
    }
    else
    {
        n = n - m;
    }
}

```

max time is $\frac{16}{2} = \frac{n}{2} = O(n)$
 min time is as $m = n = 1$ so $O(1)$

if is even

Q

```

Test (n)
{
    if (n < 5)
    {
        start
    }
    else
    {
        for (i = 0; i < n; i++)
        {
            start
        }
    }
}

```

$n = 4$
 O time $O(1)$ main
 $n = 10$
 time for loop = $O(n)$

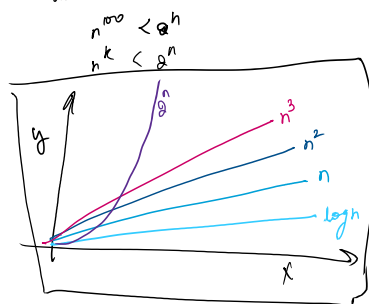
Types of Time functions or class of functions

- $O(1)$ = Constant
- $O(\log n)$ = logarithmic
- $O(n)$ = Linear
- $O(n^2)$ = Quadratic
- $O(n^3)$ = Cubic
- $O(2^n)$ = Exponential
- $O(3^n)$ =
- $O(m^n)$ =

Class of function in increasing order of derivative

$1 < \log n < \sqrt{n} < n < n \log n < n^2 < n^3 < \dots < 2^n < 3^n < n^n$

when we assume n should be large



Epsilon = 11 Asymptotic Notations

→ Notations are used to represent the simple form of functions or class of functions

→ Notations Used

O Big Oh → upper bound
 Ω Big Omega → lower bound
or Theta → Average bound

→ Big Oh definition

The function $f(n) = O(g(n))$ iff $\exists$ +ve constants c and n_0 such that $f(n) \leq c \cdot g(n) \quad \forall n \geq n_0$

e.g. $f(n) = 2n+3$

$$2n+3 \leq \frac{2}{10}n$$

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$$2n+3 \leq \frac{2}{10}n$$

easy way

$$2n+3 \leq 2n+3n$$

$$2n+3 \leq 5n$$

$$2n+3 \leq 10n \quad \text{for } n \geq 1$$

$$f(n) \leq c \cdot g(n)$$

$$\therefore f(n) = O(n)$$

Why n^2 is not valid?
It's true and it's valid?

$$2n+3 \leq 2n^2+3n^2$$

$$2n+3 \leq 5n^2$$

$$f(n) = O(n^2)$$

All values are true above n

$$1 < \log n < \sqrt{n} < n < n \log n < n^2 < n^3 < \dots < 2^n < 3^n < n^n$$

Lower bound

are valid as its upper bound

but we take nearest or closest function

Avg bound

→ Omega Notation

The function $f(n) = \Omega(g(n))$ iff $\exists$ +ve constant c and n_0 such that $f(n) \geq c \cdot g(n) \quad \forall n \geq n_0$

e.g. $f(n) = 2n+3$

$$2n+3 \geq \frac{1}{2}n \quad n \geq 1$$

$$f(n) \geq \frac{1}{2}g(n)$$

$$f(n) = \Omega(n) \quad \checkmark$$

we can say

$$2n+3 \geq 1 \times \log(n)$$

$$f(n) = \Omega(\log(n)) \quad \checkmark$$

... function because